AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

Mathematics Paper I

Time allowed: 40 minutes Marks 30

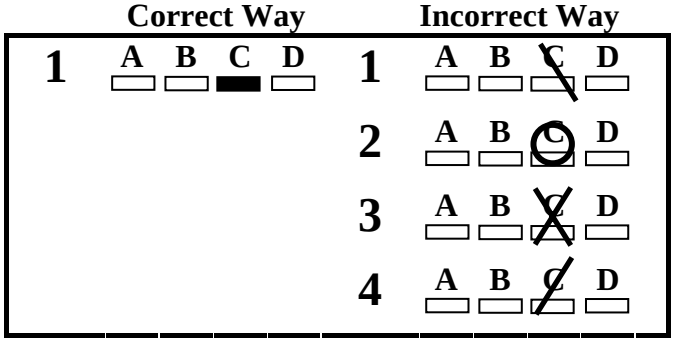
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. A set of integers is closed under

   I addition.
   II division.
   III multiplication.

   A. II only
   B. I and II only
   C. II and III only
   D. I and III only

2. The expression $\sqrt[5]{x^5 y^5}$ is equal to

   A. $x^{25} y^{25}$
   B. $x^{10} y^{10}$
   C. $x^5 y^5$
   D. xy

3. $\left(\sqrt{4}\right)^0$ is equal to

   A. 0
   B. 1
   C. 2
   D. 4

4. The real part of complex number $5i$ is

   A. -5
   B. 0
   C. 1
   D. 5

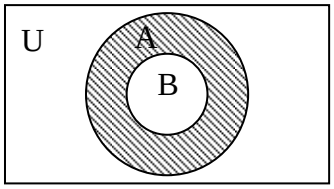
5. If $A = \{0,-1,-2\}$ and $B = \{0,1,2\}$, then $A \cap B$ is

   A. $\{\ \}$
   B. $\{\phi\}$
   C. $\{0\}$
   D. $\{0,1,2\}$

6. For any three sets *A*, *B* and *C*, $A \cap (B \cup C)$ is equal to

   A. $A \cup (B \cap C)$
   B. $A \cap (B \cap C)$
   C. $(A \cup B) \cap (A \cup C)$
   D. $(A \cap B) \cup (A \cap C)$

7. The shaded area in the given Venn diagram represents



   A. $A \cup B$
   B. $A \cap B$
   C. $B - A$
   D. $A - B$

8. If $r = \{(1, a), (2, b)\}$ is a binary relation, then domain of *r* is

   A. $\{(1,2)\}$
   B. $\{1,2\}$
   C. $\{(a,b)\}$
   D. $\{a,b\}$

9. The radius of the moon is 1738140m (approx). In scientific notation it is equal to

   A. 1.73814×10^4 m.
   B. 1.73814×10^5 m.
   C. 1.73814×10^6 m.
   D. 1.73814×10^7 m.

10. Simplified form of $\frac{a^2 - 2ab + b^2}{(a-b)(a+b)}$ is

   A. $\frac{1}{a-b}$
   B. $\frac{1}{a+b}$
   C. $\frac{a-b}{a+b}$
   D. $\frac{a+b}{a-b}$

11. The value of algebraic expression $\frac{a}{a^2+1}$ for $a=-1$ is

A. $-\frac{1}{2}$
B. 0
C. $\frac{1}{2}$
D. 1

12. If $a+b=1$ and $a-b=1$, then the value of ab is

A. 0
B. $\frac{1}{4}$
C. 1
D. 2

13. Factorized form of $-6x^2-10x$ is

A. $2x(3x-5)$
B. $-2x(3x-5)$
C. $-2x(3x+5)$
D. $2x(3x+5)$

14. The remainder of $(x+1)\div(x-1)$ is

A. -1
B. 0
C. 1
D. 2

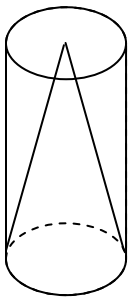
15. The fourth proportional of $\frac{1}{2}, \frac{3}{2}$ and $\frac{2}{3}$

A. $\frac{2}{9}$
B. $\frac{4}{9}$
C. $\frac{1}{2}$
D. 2

16. Twelve workers were hired to bind 1000 books in four days. Before starting the work, four more workers were hired to help complete the task. The estimated time for completion of the task is

A. two days.
B. three days.
C. four days.
D. six days.

17. For equal radii, the ratio of volumes of a cone and a cylinder is 1: 3. If volume of the cylinder is $6cm^3$, then the volume of cone of same radius is



A. $2\,cm^3$.
B. $9\,cm^3$.
C. $18\,cm^3$.
D. $27\ cm^3$.

18. The matrix $\begin{bmatrix}1 & 1 & -1\end{bmatrix}$ is a

A. column matrix.
B. scalar matrix.
C. row matrix.
D. unit matrix.

19. The additive identity of every 2×2 matrix is

A. $\begin{bmatrix}1 & 0\\ 1 & 0\end{bmatrix}$
B. $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$
C. $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$
D. $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

20. Which one of the following is a skew-symmetric matrix?

A. $\begin{bmatrix}0 & 2\\ 2 & 0\end{bmatrix}$
B. $\begin{bmatrix}0 & 2\\ -2 & 0\end{bmatrix}$
C. $\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$
D. $\begin{bmatrix}2 & 2\\ 0 & 0\end{bmatrix}$

21. The ages (in years) of 9 children in a group are 7, 7, 8, 9, 6, 8, 5, 8 and 10. The modal age is

A. 6
B. 7
C. 8
D. 9

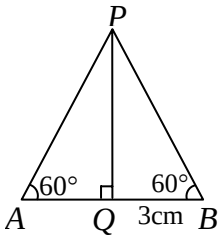
22. The marks obtained by 8 students in a class are 40, 60, 55, 65, 73, 85, 68 and 70. The range of the marks is

A. 85
B. 73
C. 65
D. 45

23. Ahmed arranged the number of books owned by his friends in ascending order as 10, 12, 15, x, 17, 18. If the median of the data is 16, then the value of x is

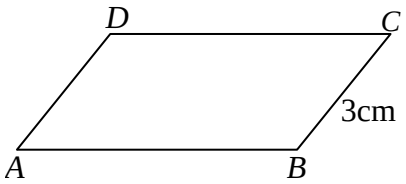
A. 15
B. 15.5
C. 16
D. 17

24. In the given diagram $m\overline{AB}$ is



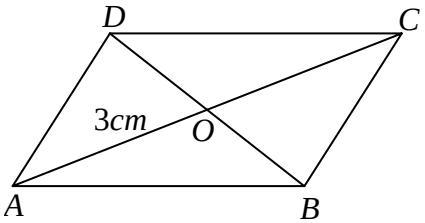
A. $3cm$.
B. $4cm$.
C. $6cm$.
D. $7cm$.

25. If the perimeter of the given parallelogram is $16cm$ and $m\overline{BC} = 3cm$, then $m\overline{AB}$ is



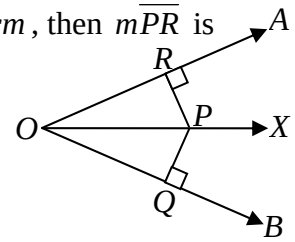
A. $3cm$.
B. $4cm$.
C. $5cm$.
D. $6cm$.

26. In the given parallelogram $m\overline{AO} = 3cm$. $m\overline{CO}$ is



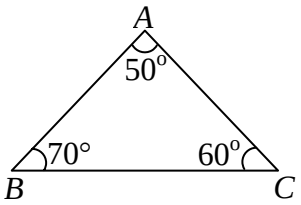
A. $3cm$.
B. $6cm$.
C. $9cm$.
D. $12cm$.

27. In the given figure, $\overrightarrow{OX}$ is angle bisector of $\angle AOB$. If $m\overline{PQ} = 3cm$, then $m\overline{PR}$ is



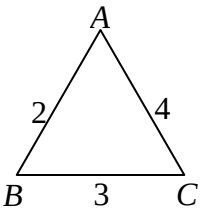
A. $2cm.$
B. $3cm.$
C. $4cm.$
D. $6cm.$

28. For the given triangle ΔABC, which one of the following statements is true?



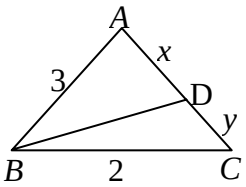
A. $m\overline{AC} > m\overline{BC}$
B. $m\overline{AB} > m\overline{AC}$
C. $m\overline{BC} > m\overline{AB}$
D. $m\overline{BC} > m\overline{AC}$

29. For the given ΔABC, which one of the following is true?



A. $m\angle C < m\angle A < m\angle B$
B. $m\angle B < m\angle A < m\angle C$
C. $m\angle A < m\angle B < m\angle C$
D. $m\angle C < m\angle B < m\angle A$

30. If $\overline{BD}$ is the internal bisector of $\angle ABC$, then the ratio of $x : y$ is



A. 1 : 2
B. 3 : 1
C. 2 : 3
D. 3 : 2

END OF PAPER

Please use this page for rough work